

Python Variable Introduction

```
In [1]: v = 5  
v
```

```
Out[1]: 5
```

```
In [4]: 5 = v    # Incorrect manner to declare
```

```
Cell In[4], line 1  
    5=v  
    ^  
SyntaxError: cannot assign to literal here. Maybe you meant '==' instead of '='?
```

```
In [6]: v = 5  
v
```

```
Out[6]: 5
```

```
In [7]: 6
```

```
Out[7]: 6
```

Rules to define Python Variable

```
In [8]: v1 = 45
```

```
In [9]: V1
```

```
-----  
NameError                                Traceback (most recent call last)  
Cell In[9], line 1  
----> 1 V1  
  
NameError: name 'V1' is not defined
```

```
In [10]: v1
```

```
Out[10]: 45
```

```
In [11]: 1v = 50    # Varialbe can't start with a number
```

```
Cell In[11], line 1  
    1v = 50  
    ^  
SyntaxError: invalid decimal literal
```

```
In [12]: 23a = 56  
23a
```

```
Cell In[12], line 1
    23a = 56
    ^
SyntaxError: invalid decimal literal
```

```
In [14]: a23 = 56      # Variable can start with Alphabets (A - Z) or (a - z)
a23
```

Out[14]: 56

```
In [ ]: mit@ = 45      # Variables don't contain special characters except '_'
```

```
Cell In[15], line 1
    mit@ = 45
    ^
SyntaxError: invalid syntax
```

```
In [16]: nit_ = 45      # Underscore can be used in declaration of a variable
nit_
```

Out[16]: 45

```
In [17]: nit% = 48
nit%
```

```
Cell In[17], line 1
    nit% = 48
    ^
SyntaxError: invalid syntax
```

```
In [18]: import keyword
keyword.kwlist
```

```
Out[18]: ['False',
          'None',
          'True',
          'and',
          'as',
          'assert',
          'async',
          'await',
          'break',
          'class',
          'continue',
          'def',
          'del',
          'elif',
          'else',
          'except',
          'finally',
          'for',
          'from',
          'global',
          'if',
          'import',
          'in',
          'is',
          'lambda',
          'nonlocal',
          'not',
          'or',
          'pass',
          'raise',
          'return',
          'try',
          'while',
          'with',
          'yield']
```

```
In [19]: len(keyword.kwlist)
```

```
Out[19]: 35
```

```
In [20]: if = 56      # Keyword can't be used as a Variable
         if
```

```
Cell In[20], line 1
      if = 56
      ^
SyntaxError: invalid syntax
```

```
In [21]: IF = 56
         IF
```

```
Out[21]: 56
```

```
In [22]: for = 67
         for
```

```
Cell In[22], line 1
    for = 67
      ^
SyntaxError: invalid syntax
```

```
In [23]: FOR = 67
        FOR
```

Out[23]: 67

```
In [1]: age = 21  # Case sensitive
```

```
In [2]: age
```

Out[2]: 21

```
In [3]: Age = 25
```

```
In [4]: Age
```

Out[4]: 25

```
In [5]: AGE = 30
```

```
In [6]: AGE
```

Out[6]: 30

```
In [7]: age
        Age
        AGE
```

Out[7]: 30

Print Function

```
In [24]: 10 + 5
        10 - 5
        10 * 5
        10 / 5    # Float division
        10 // 5   # Int division
```

Out[24]: 2

```
In [25]: print(10 + 5)
        print( 10 -5)
        print(10 * 5)
        print(10 / 5)
        print(10// 5)
```

15

5

50

2.0

2

```
In [26]: a = 56  
         print(a)
```

56

```
In [ ]: #Example 1: Integer Variable
```

```
In [1]: age = 25  
        print(age)
```

25

```
In [ ]: #Example 2: String Variable
```

```
In [2]: name = 'Alice'  
        print(name)
```

Alice

```
In [ ]: #Example 3: Float Variable
```

```
In [3]: price = 19.99  
        print(price)
```

19.99

```
In [ ]: #Example 4: Boolean Variable
```

```
In [4]: is_active = True  
        print(is_active)
```

True

1. Storing and Printing a Value

```
In [5]: x = 10  
        print(x)
```

10

2. Using Variables in Expressions

```
In [6]: a = 5  
        b = 3  
        result = a + b  
        print(result)
```

8

3. Changing the value of a Variable

```
In [7]: score = 50  
        print(score)
```

50

```
In [8]: score = 100  
print(score)
```

100

4. Concatenating Strings

```
In [9]: first_name = "John"  
last_name = "Doe"  
full_name = first_name + " " + last_name  
print(full_name)
```

John Doe

5. Using Variables in a Calculation

```
In [10]: length = 10  
width = 5  
area = length * width  
print(area)
```

50

6. Reassigning Values to Variables

```
In [11]: x = 10  
print(x)
```

10

```
In [12]: x = 20  
print(x)
```

20

Changing the Variable Declaration

```
In [5]: #Variables in Python  
first_name = 'Priya Darshani'  
last_name = 'Sahu'  
country = 'India'  
city = 'Berhampur'  
age = 23  
is_married = False  
skills = ['HTML' , 'CSS' , 'JS' , 'React' , 'Python']  
person_info = {  
    'firstname': 'Shreyash',  
    'lastname': 'Sahu',  
    'country': 'India',  
    'city': 'Bhubaneswar'  
}  
  
#Printing the values stored in the variables  
  
print('First name:', first_name)  
print('First name length:', len(first_name))
```

```
print('Last name:', last_name)
print('last name length:', len(last_name))
print('Country:', country)
print('City:', city)
print('Age:', age)
print('Married:', is_married)
print('Skills:', skills)
print('Person information:', person_info)
```

First name: Priya Darshani

First name length: 14

Last name: Sahu

last name length: 4

Country: India

City: Berhampur

Age: 23

Married: False

Skills: ['HTML', 'CSS', 'JS', 'React', 'Python']

Person information: {'firstname': 'Shreyash', 'lastname': 'Sahu', 'country': 'India', 'city': 'Bhubaneswar'}

In [4]: *#Declaring multiple variables in one line*

```
first_name, last_name, country, age, is_married = 'Priya Darshani', 'Sahu', 'India', 23, False

print(first_name, last_name, country, age, is_married)
print('First name:', first_name)
print('Last name:', last_name)
print('Country:', country)
print('Age:', age)
print('Married:', is_married)
```

Priya Darshani Sahu India 23 False

First name: Priya Darshani

Last name: Sahu

Country: India

Age: 23

Married: False

In []: